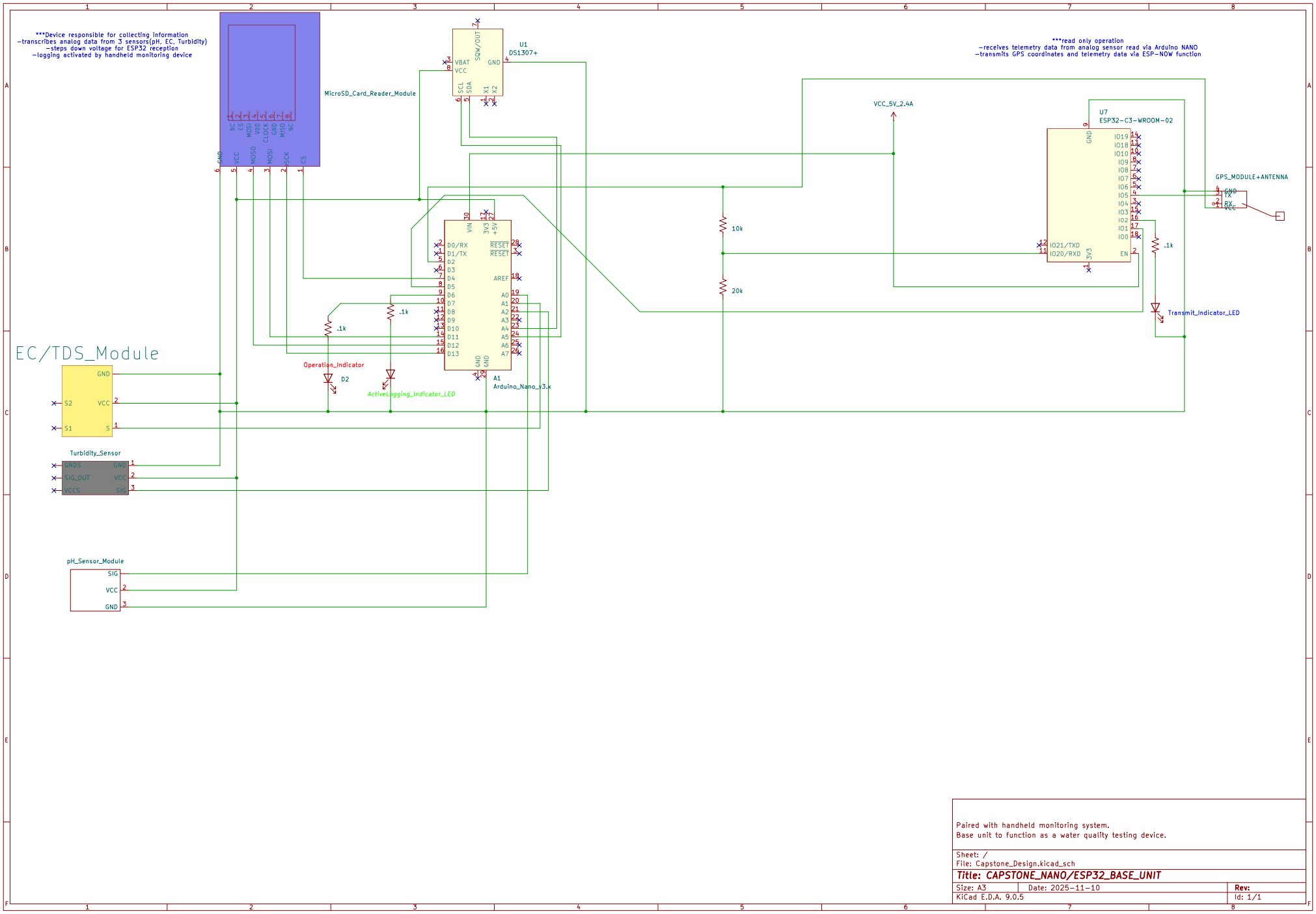
***Device responsible for collecting information
−transcribes analog data from 3 sensors(pH, EC, Turbidity)
−steps down voltage for ESP32 reception
−logging activated by handheld monitoring device
MicroSD_Card_Reader_Module
U1
DS1307+
***read only operation
−receives telemetry data from analog sensor read via Arduino NANO
−transmits GPS coordinates and telemetry data via ESP−NOW function
VCC_5V_2.4A
U7
ESP32−C3−WROOM−02
GPS_MODULE+ANTENNA
10k
20k
.1k
.1k
.1k
Transmit_Indicator_LED
Operation_Indicator
D2
ActiveLogging_Indicator_LED
A1
Arduino_Nano_v3.x
EC/TDS_Module
Turbidity_Sensor
pH_Sensor_Module
Paired with handheld monitoring system.
Base unit to function as a water quality testing device.
Sheet: /
File: Capstone_Design.kicad_sch
Title: CAPSTONE_NANO/ESP32_BASE_UNIT
Size: A3
Date: 2025−11−10
Rev:
KiCad E.D.A. 9.0.5
Id: 1/1